MENDSPEECH REVISED LEARNING SYSTEM

MendSpeech Complete Learning and Research Roadmap

A 56 day systems and research curriculum for selective semantic speech restoration, real time streaming ASR, calibrated repair decisions, boundary matched reconstruction, and direct audio repair comparison.

> Eight weeks is the minimum structure. A realistic execution range is eight to ten weeks and roughly 150 to 185 focused hours.

1. Final Project

> **MendSpeech** is a research grade speech restoration system that preserves trustworthy original speech, inspects uncertain regions, repairs only when the inferred content is reliable, and abstains when reconstruction would be guesswork.

```
Damaged speech
     |
SpeechDamageBench or live input
     |
Streaming FastConformer
     |
Alignment + calibrated uncertainty
     |
Preserve / Inspect / Repair / Abstain
     |
     +--> MendSpeech V1: speaker conditioned TTS + boundary matching
     |
     +--> Direct audio baseline: pretrained latent or codec inpainting
     |
Shared evaluation: intelligibility, preservation, speaker similarity, latency, RTF, seam quality
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2. What the Revision Changes

- The cascaded ASR to text to TTS path is now an explicit real time baseline, not a claim of frontier restoration.
- Week 7 adds short time energy matching, local loudness matching, room tone handling, and equal power crossfades before stitching.
- SpeechDamageBench becomes an installable, deterministic, versioned package with seed controlled corruptions.
- Week 8 compares the cascaded baseline against one reproducible pretrained direct latent or codec audio inpainting system.
- The timeline is treated honestly: eight weeks is the structure, while eight to ten weeks is a more realistic completion window.

3. Eight Week Structure

Week	Focus	Project milestone
1	Audio, Degradation, and Measurement Foundations	Build the audio laboratory and make SpeechDamageBench a deterministic standalone package.
2	ASR, CTC, Confidence, and Repair Localization	Build the recognition and uncertainty layer, plus a reusable Modal execution path.
3	Conformer From First Principles	

		Implement the core encoder pieces so model behavior is not a black box.
4	FastConformer and Efficient Encoder Behavior	Measure why FastConformer is efficient and freeze a reproducible baseline.
5	Streaming, Cache Aware Inference, and Adaptive Context	Turn the recognizer into a real time system and test uncertainty guided context spending.
6	Robustness, Fine Tuning, RNNT, and Calibration	Adapt the recognizer to damaged speech while learning training and calibration discipline.
7	TTS, Speaker Preservation, and Boundary Matched Reconstruction	Build MendSpeech V1 as a cascaded selective repair baseline with explicit seam diagnostics.
8	Research Capstone: Cascaded Versus Direct Repair	Freeze the benchmark, run controlled ablations, compare architectures, and publish a reproducible result.

4. Realistic Workload

Phase	Expected effort	Main risk
Weeks 1 to 3	35 to 45 hours	Low. Mostly NumPy, PyTorch, audio analysis, and small model components.
Weeks 4 to 5	45 to 55 hours	High. Framework configuration, cache state, streaming masks, GPU timing, and deployment details.
Week 6	25 to 35 hours	Medium. Data splits, training stability, memory, calibration, and evaluation discipline.
Weeks 7 to 8	40 to 50 hours	Medium to high. Speaker conditioned synthesis, seam debugging, direct audio baseline integration, and research reporting.
Total	**150 to 185 hours**	**Do not sacrifice understanding to preserve a calendar date.**

5. Daily Operating Protocol

Read

25 minutes. Use the primary paper or framework documentation only for the concept needed today.

Build

65 minutes. Implement, modify, or run one controlled experiment. Change one major variable at a time.

Notebook

20 minutes. Hypothesis, method, result, surprise, limitation, next question.

Commit and explain

10 minutes. Save evidence and explain the day without reading notes.

Important: two hours is the core session, not a promise that every debugging problem fits inside two hours. A task is complete only when its evidence and completion check are satisfied.

6. Hardware Strategy

- Weeks 1 to 3: local CPU for most work. Use an L4 only when scaling model runs.
- Weeks 4 to 8: Modal L4 is the default GPU for reproducible inference, streaming, training, and benchmarks.
- Use a larger GPU only when memory blocks the planned experiment after batch size, mixed precision, and gradient accumulation have been considered.
- Keep hardware fixed inside any latency or throughput comparison.

7. Research Questions

1. Can selective repair improve intelligibility while retaining more original speech than full resynthesis?
2. Can calibrated uncertainty allocate streaming context more efficiently than a fixed lookahead policy?
3. Can boundary matching reduce the audible seams created by short span TTS reconstruction?
4. When does the cascaded real time path outperform or underperform a direct latent or codec audio inpainting baseline?

8. Definition of Done

- SpeechDamageBench is independently installable, deterministic, and versioned.
- MendSpeech exposes streaming text, uncertainty, context state, preserve or repair intervals, and latency.
- MendSpeech V1 can selectively reconstruct spans with measured boundary matching instead of naive waveform replacement.
- The system abstains when content is too uncertain to reconstruct responsibly.
- The capstone compares raw damaged audio, full resynthesis, cascaded selective repair, and one direct audio repair baseline on the same frozen cases.
- Every major claim in the report points to a saved result, table, or figure.
- A fresh environment can reproduce at least one benchmark from documented commands.

9. Resource Spine

- PyTorch and TorchAudio documentation for signal processing and tensor work.
- CTC primary material and framework decoding examples.
- Conformer primary paper and a mature implementation.
- FastConformer primary paper and NVIDIA NeMo model documentation.
- Cache aware or stateful Conformer primary material and NeMo streaming examples.
- RNNT references and NeMo training documentation.
- FastSpeech 2, HiFi GAN, and VITS for the cascaded synthesis baseline.
- One reproducible pretrained direct latent or codec audio inpainting system for Week 8 architecture comparison.